

Critical Values for the KS, CvM, and AD Goodness-of-Fit Tests for the Log-Normal Distribution with Estimated Parameters

Chanry Shuai
Georgia Institute of Technology

Abstract

Critical values for standard EDF goodness-of-fit tests (KS, CvM, AD) are invalid when parameters are estimated—a problem previously addressed for the normal distribution. This study extends the analysis to the log-normal case, generating new critical values via Monte Carlo simulation. The results reveal a striking finding: the critical values for the log-normal test are nearly identical to the established values for the normal test. This suggests the null distribution is primarily dictated by the parameter estimation process rather than the underlying distribution family, leading to a crucial practical conclusion: widely available tables for the normal test serve as a highly accurate approximation for log-normal goodness-of-fit tests.

1. INTRODUCTION

The fundamental task in a goodness-of-fit analysis is to test the null hypothesis (H_0) that a given random sample, $x_1, x_2, \dots, x_n$, originates from a population of a specific distribution function, $F(x)$. While several methods exist for this purpose, this paper focuses on a class of goodness-of-fit statistics known as *Empirical Distribution Function (EDF) statistics*. These tests are founded on the comparison between the hypothesized distribution function, $F(x)$, and the empirical distribution function, $F_n(x)$, derived from the sample data. Within the broad family of EDF statistics, this study will focus on three widely recognized and crucial tests: the **Kolmogorov-Smirnov (KS)**, the **Cramér-von Mises (CvM)**, and the **Anderson-Darling (AD)** tests.

For continuous distributions, it is well-established that EDF statistics generally provide more powerful tests than the traditional chi-square (χ^2) test. However, their application has historically been limited due to certain disadvantages, one of which is when parameters of the distribution are not known and must be estimated from the sample.

This study focuses on generating critical value tables for the KS, AD, and CvM test statistics under a *log-normal* null hypothesis with *estimated parameters*. The paper is structured as follows: Section 2 provides a background on the EDF statistics. Section 3 details the Monte Carlo simulation procedure used to generate the critical values. Section 4 presents the resulting tables and

a validation of the findings against established benchmarks in literature. Finally, Section 5 offers concluding remarks.

1.1. THE LOG-NORMAL DISTRIBUTION

As noted above, the distribution selected for this investigation is the log-normal distribution. A continuous random variable $X \sim \text{LogNormal}(\mu, \sigma)$ is said to be log-normally distributed if its natural logarithm, $Y = \ln(X)$, follows a normal distribution. This distribution is widely used to model *non-negative, right-skewed, tail-heavy* data that arise in a variety of disciplines, from finance and economics to engineering and the biological sciences (Mathworks, n.d., Wolfram, n.d.).

(Wolfram, n.d.) The probability density function (PDF) of the log-normal distribution is given by:

$$f(x | \mu, \sigma) = \frac{1}{x\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln(x) - \mu)^2}{2\sigma^2}\right), \quad \text{for } x > 0$$

where μ is the mean and σ is the standard deviation

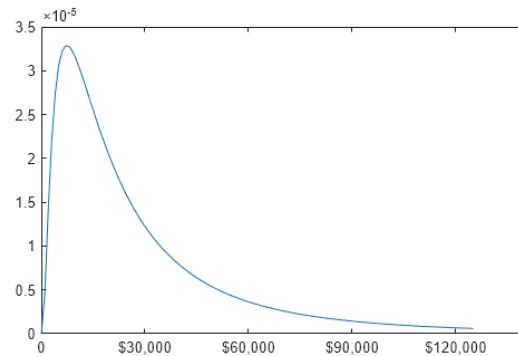


Figure 1. Example of a log-normal probability density function, illustrating its characteristic right skew (MathWorks, n.d.).

The cumulative distribution function (CDF) can be expressed using the standard normal CDF, Φ :

$$F(x | \mu, \sigma) = \Phi \left(\frac{\ln(x) - \mu}{\sigma} \right)$$

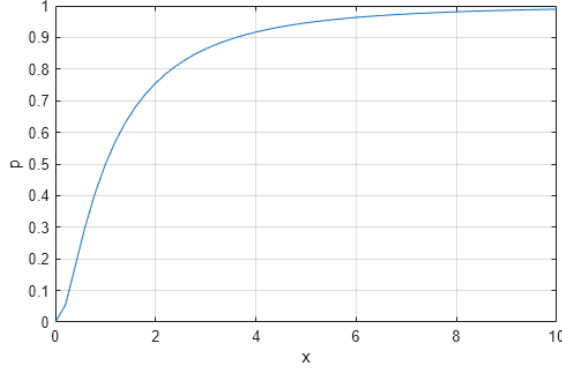


Figure 2. Example of a log-normal cumulative distribution function (MathWorks, n.d.).

As illustrated in the figures above, the log-normal distribution is bounded at zero and is characterized by a pronounced right skew and a long right tail. This visual characteristic is confirmed by its theoretical skewness:

$$\gamma_1 = \sqrt{e^{\sigma^2} - 1} (2 + e^{\sigma^2})$$

The skewness of a $\text{LogNormal}(0,1) \approx 6.183$ (in contrast, the skewness of any normal distribution = 0). The log-normal distribution's skewness is always positive for $\sigma > 0$, thus its inherent asymmetry. The degree of skewness and the weight of the tail are determined by σ .

The log-normal distribution was selected for this study by the following key factors:

- while frequently applied, there is a scarcity of comprehensive tables of critical values for EDF tests, in contrast to more common distributions, normal, exponential, etc.
- its skewness and tail-heavy nature make it an excellent candidate for examining the CvM test and, especially, the AD test.

2. BACKGROUND ON EDF GOODNESS-OF-FIT STATISTICS

2.1. THE KOLMOGOROV-SMIRNOV (KS) TEST

The Kolmogorov-Smirnov (KS) test measures the maximum vertical difference between the hypothesized cumulative distribution function (CDF), $F^*(x)$, and the empirical distribution

function (EDF), $F_n(x)$, derived from the sample data. The test statistic, D , is formally defined as the supremum of this difference across all values of x :

$$D = \sup_x |F^*(x) - F_n(x)|$$

For practical computation from a set of ordered data points ($x_1 < x_2 < \dots < x_n$), the statistic, denoted D_n , is calculated using the following components:

$$D_n^+ = \max_i [i/n - F^*(x_i)]$$

$$D_n^- = \max_i [F^*(x_i) - (i-1)/n]$$

$$D_n = \max(D_n^+, D_n^-)$$

LILLIEFORS ADAPTATION:

A critical issue with this standard procedure arises when parameters are unknown, a problem famously addressed by Lilliefors (1967). Using the standard tables in this scenario produces what Lilliefors describes as an "*extremely conservative*" test, meaning the actual probability of a TYPE I error is in fact significantly smaller than the level selected. To address this for the normal distribution, Lilliefors proposed a modification that has become standard practice. He constructed a new table of critical values specifically for the case where the mean and variance are estimated. The procedure retains the original statistic D , but with changes made in the definition of the hypothesized distribution, $F^*(x)$, where μ and σ^2 are replaced by their *sample* estimates. Using Monte Carlo simulation, Lilliefors generated the appropriate critical values for this statistic and found them to be substantially lower—approximately **two-thirds** of the standard KS values. This confirms that a different table of critical values is essential for an accurate test.

2.2. THE CRAMÉR-VON MISES (CVM) TEST

An alternative to the KS test is the Cramér-von Mises (CvM) statistic, W^2 , which evaluates the overall fit by considering squared difference between the empirical and hypothesized distributions, rather than just the single point of maximum distance. The formal definition of the statistic is:

$$W_n^2 = n \int_{-\infty}^{\infty} [F_n(x) - F^*(x)]^2 dF^*(x)$$

For computational purposes, Stephens (1974) provides a more direct formula based on the transformed data points, $z_i = F^*(x_i)$:

$$W^2 = \sum_{i=1}^n \left(z_i - \frac{2i-1}{2n} \right)^2 + \frac{1}{12n}$$

2.3. THE ANDERSON-DARLING (AD) TEST

The Anderson-Darling (AD) test, similarly, is a *quadratic* EDF statistic that can be seen as a modification of CvM. Its defining characteristic is that it assigns even more weight to observations in the tails of the distribution, making it particularly sensitive to discrepancies in these regions. It introduces an additional weighting factor as shown in its formal definition:

$$A^2 = n \int_{-\infty}^{\infty} \frac{(F_n(x) - F^*(x))^2}{F^*(x)(1 - F^*(x))} dF^*(x)$$

The weighting term, $[F^*(x)(1 - F^*(x))]^{-1}$ places greater emphasis on the smallest and largest values of the CDF, a key difference from the KS and CvM tests. For computational purposes from the transformed data points z_i , the statistic is calculated as: (Stephens, 1974)

$$A^2 = - \frac{\sum_{i=1}^n (2i-1)[\ln(z_i) + \ln(1 - z_{n+1-i})]}{n} - n$$

Due to its sensitivity, the AD test is often considered one of the most powerful EDF tests available.

**notes on z_i transformation and uniformity:*

EDF statistics is typically “distribution-free”. However, this property *only holds true when the parameters of the hypothesized distribution are fully specified in advance* (Stephens, 1974, “Case 0”). In this case, the probability integral transform $z_i = F(x_i | \mu, \sigma)$ is uniformly distributed, regardless of the original distribution F . Since the test statistics are calculated from these uniform values, their null distributions are independent of F .

However, in the more common and practical case (Evans et al., 2008, Stephens, 1974, “Case 3”), as well as in the Monte Carlo case of this study, parameters must be **estimated from the data**, the tests are no longer distribution-free. The transform $z_i = F(x_i | \bar{x}, s)$ uses sample estimates, causing the resulting z_i values to be “super-uniform,” or more regular than a truly random sample. *The nature of this deviation from uniformity depends on the family of the underlying distribution being tested.* This

phenomenon, first rigorously demonstrated for the normal distribution by Lilliefors (1967), necessitates the generation of specific critical value tables for certain distributional test cases. This paper addresses the case of a log-normal distribution.

3. MONTE CARLO SIMULATION APPROACH

This study follows a Monte Carlo simulation approach, similar to Razali (2011) who conducted the power analysis of various specific distributions. The objective is to estimate the null distribution and corresponding critical values for the KS, CvM, and AD test statistics.

The null (H_0) and alternative (H_1) hypotheses:

H_0 : The sample data are drawn from a log-normal distribution.

H_1 : The sample data are not drawn from a log-normal distribution.

The critical values of interests are **0.75, 0.90, 0.95, and 0.99 quantiles**, which are dependent on the sample size, **n**. Therefore, the simulation was performed for a range of sample sizes: **n** = {10, 15, 20, 25, 30, 40, 50, 100, 200, 300, 400, 500, 1000, 1500, 2000}.

(Razali, 2011) For each sample size, the simulation procedure was executed as follows:

1. A random sample of size **n** was drawn from a standard log-normal distribution (with $\mu=0$ and $\sigma=1$).

2. Estimate the parameters of the log-normal distribution: take the natural logarithm of the sample data and calculate the sample mean ($\bar{y}$) and sample standard deviation (s_y).

3. With estimated parameters, transform sample data to a set of z_i values.

4. The KS (D), CvM (W^2), and AD (A^2) statistics were calculated from the z_i values.

5. Steps 1-4 were repeated 10,000 times, creating an empirical null distribution.

6. The 10,000 simulated values were sorted. The critical values for the desired quantiles were determined from this sorted list for each test statistics.

This entire process was repeated for each sample size **n** to generate the final tables.

**note: this simulation utilizes the *lognormal* function from Python’s NumPy library to produce the random variates.*

Table 1: Condensed Critical Values for KS, CvM, and AD Tests by Sample Size and Significance Level

Test	Sample Size (n)	$\alpha = 0.25$	$\alpha = 0.10$	$\alpha = 0.05$	$\alpha = 0.01$
KS	10	0.2093	0.2421	0.2631	0.3037
	50	0.0990	0.1149	0.1248	0.1450
	100	0.0707	0.0816	0.0888	0.1034
	2000	0.0161	0.0186	0.0203	0.0235
CvM	10	0.0719	0.1003	0.1223	0.1716
	50	0.0738	0.1033	0.1263	0.1818
	100	0.0736	0.1041	0.1242	0.1731
	2000	0.0743	0.1043	0.1266	0.1826
AD	10	0.4364	0.5832	0.6993	0.9594
	50	0.4645	0.6329	0.7439	1.0342
	100	0.4679	0.6305	0.7443	0.9950
	2000	0.4704	0.6305	0.7514	1.0434

4. RESULTS AND ANALYSIS

4.1. SIMULATED CRITICAL VALUE TABLES

For KS, CvM, and AD tests for a log-normal null hypothesis with estimated parameters, Table 1 presents a condensed version of these critical values for selected sample sizes. The full, comprehensive tables for all simulated sample sizes are provided in the Appendix (Tables A1, A2, and A3).

The values in the Table 1 exhibit clear trends. For all tests, critical values decrease as sample size n increases. The CvM and AD statistics stabilize much more rapidly than the KS statistic, suggesting a fast convergence.

4.2. VALIDATION: COMPARISON WITH STANDARD BENCHMARKS

To validate the necessity and uniqueness of the new tables, the simulated critical values are compared against standard benchmarks from the literature.

4.2.1. COMPARISON FOR THE KS TEST

In his comprehensive 1974 paper, Stephens discusses five cases regarding the knowledge of the distribution $F(x)$. For the purpose of this validation, we are interested in two of Stephens' cases (see also *notes on z_i transformation and uniformity*):

Case 0: $F(x)$ is continuous and completely specified.

Case 3: $F(x)$ is the normal distribution with both μ and σ^2 unknown and estimated from the sample.

Specifically, the critical values for the KS test calculated in this study are compared against three standard benchmarks:

- Fully Specified Case (Stephens, 1974, Table 1.0): standard, distribution-free (Case 0).
- Normal Test Case (Stephens, 1974, Table 1.3): estimated parameters (Case 3).
- Normal Test Case (Lilliefors, 1967): original Monte Carlo critical values with estimated parameters (Case 3).

Table 2 below presents this comparison for a representative sample size of $n=50$. The benchmark critical values were derived as follows:

1. Fully Specified Case (Stephens, 1974, Table 1.0): reverse-engineered the formula from Stephens' case 0 "*Illustration 1*" (Section 3.3):

$$D = D^*/(\sqrt{n} + 0.12 + 0.11/\sqrt{n})$$

for example,

$$\alpha = 0.10 : D = 1.224/(\sqrt{50} + 0.12 + 0.11/\sqrt{50}) = \mathbf{0.170}$$

2. Estimated Normal Test Case (Stephens, 1974, Table 1.3): reverse-engineered the formula analogous to Stephens' case 3 "*Illustration 2*":

$$D = D^*/(\sqrt{n} - 0.01 + 0.85/\sqrt{n})$$

for example,

$$\alpha = 0.10 : D = 0.819/(\sqrt{50} - 0.01 + 0.85/\sqrt{50}) = \mathbf{0.114}$$

3. Estimated Normal Test Case (Lilliefors, 1967): For $n > 30$, Lilliefors provides the formula:

$$D = \text{Constant}/\sqrt{n}$$

for example,

$$\alpha = 0.10 : D = 0.805/\sqrt{50} = \mathbf{0.114}$$

Table 2: Comparison of Log-Normal with Standard Benchmarks for the KS Test (n=50)

Test Case	$\alpha = 0.10$	$\alpha = 0.05$	$\alpha = 0.01$
1. Fully Specified Case (Stephens 1.0)	0.170	0.188	0.226
2. Normal Test Case (Stephens 1.3)	0.114	0.125	0.144
3. Normal Test Case (Lilliefors)	0.114	0.125	0.146
4. Log-Normal Test (This Study)	0.115	0.125	0.145
Ratio			
Log-Normal / Fully Specified	0.676	0.665	0.642
Log-Normal / Normal Test (Stephens)	1.009	1.000	1.007
Log-Normal / Normal Test (Lilliefors)	1.009	1.000	0.993

These results lead to several discoveries:

- the comparison between 4. *Log-Normal Test* and 1. *Fully Specified Case* confirms that the critical values for the log-normal test with estimated parameters are only **64-68%** of the standard, distribution-free values. This aligns with the observation by Lilliefors (1967) that his Monte Carlo critical values for the normal test were "*in most cases approximately two-thirds the standard values*," providing a strong validation of our simulation's accuracy. Thus, using standard, fully-specified tables for this test would be incorrect and would lead to a significant loss of power.
- two benchmarks — Stephens and Lilliefors, 2. and 3. *Normal Test Case* are consistent, validating their methods for the normal distribution with estimated parameters.
- the comparison between 4. *Log-Normal Test* and 2. and 3. *Normal Test Case* addresses a more practical question. For the KS test, the critical values for the log-normal test are nearly identical to those for the normal test, with ratios very close to 1.0. This suggests that, **for the KS test**, using the widely available Lilliefors tables

might be a reasonable approximation when testing for a log-normal distribution.

This paper will then investigate whether this same observation holds for the more powerful CvM and AD tests.

4.2.2. COMPARISON FOR THE CVM TEST

The same comparison was performed for the CvM (W^2) statistic. Table 3 compares the simulated critical values for the log-normal test against the standard benchmarks for the fully specified case (Case 0) and the normal test with estimated parameters (Case 3) from Stephens (1974).

Similar to the KS case, the benchmark critical values for n=50 were derived as follows:

1. Fully Specified Case (Stephens, 1974, Table 1.0): W^2 is calculated by algebraically solving Stephens' modification formula: $W^{2*} = (W^2 - 0.4/n + 0.6/n^2)(1.0 + 1.0/n)$.

for example,

$$\alpha = 0.10 : W^2 = (0.347/1.02) + 0.00776 = \mathbf{0.348}$$

Table 3: Comparison of Log-Normal with Standard Benchmarks for the CvM Test (n=50)

Test Case	$\alpha = 0.10$	$\alpha = 0.05$	$\alpha = 0.01$
1. Fully Specified Case (Stephens 1.0)	0.348	0.460	0.736
2. Normal Test Case (Stephens 1.3)	0.103	0.125	0.176
3. Log-Normal Test (This Study)	0.103	0.126	0.182
Ratio			
Log-Normal / Fully Specified	0.296	0.274	0.247
Log-Normal / Normal Test	1.000	1.008	1.034

Table 4: Comparison of Log-Normal with Standard Benchmarks for the AD Test (n=50)

Test Case	$\alpha = 0.10$	$\alpha = 0.05$	$\alpha = 0.01$
1. Fully Specified Case (Stephens 1.0)	1.933	2.492	3.857
2. Normal Test Case (Stephens 1.3)	0.613	0.736	1.021
3. Log-Normal Test (This Study)	0.633	0.744	1.034
Ratio			
Log-Normal / Fully Specified	0.327	0.299	0.268
Log-Normal / Normal Test	1.033	1.011	1.013

2. Estimated Normal Test Case (Stephens, 1974, Table 1.3): W^2 is found by reverse-engineering the formula for Case 3: $W^{2*} = W^2(1.0 + 0.5/n)$. for example,
 $\alpha = 0.10$: $W^2 = 0.104/1.01 = 0.103$

Findings from the comparison for the CvM statistic:

- the ratio of the *Log-Normal Test* values to the *Fully Specified Normal* values is only **0.25-0.30**, even more dramatic than the KS test. This extremely large discrepancy provides stronger evidence that using standard, distribution-free tables for this test would be highly inappropriate.
- the critical values for the *Log-Normal Test* are again nearly identical to those for the *Normal Test Case*, with ratios very close to 1.0 for $n=50$. Therefore, the null distribution for a log-normal test with estimated parameters is similar to that of a normal test, for small sample sizes.

*note: published tables for the CvM statistic for the normal distribution with sample sizes >1000 are not widely available, as the asymptotic distribution is typically used for such large samples. We will rely on asymptotic critical values from Stephens (1970).

4.2.3. COMPARISON FOR THE AD TEST

Finally, the same comparison was performed for the Anderson-Darling (A^2) statistic. Table 4 compares the simulated critical values for the log-normal test against the standard benchmarks for the fully specified case (Case 0) and the normal test with estimated parameters (Case 3) from Stephens (1974)

The benchmark critical values for $n=50$ were derived as follows:

1. Fully Specified Case (Stephens, 1974, Table 1.0): Stephens notes that for A^2 , no modification

is required for $n \geq 5$. Therefore, the sample-size-specific critical value is the same as the published asymptotic value. For example, $\alpha = 0.10$:
 $A^2 = 1.933$

2. Estimated Normal Test Case (Stephens, 1974, Table 1.3): A^2 is found by reverse-engineering the formula for Case 3:

$$A^{2*} = A^2(1.0 + 4/n - 25/n^2)$$

for example,

$$\alpha = 0.10 : A^2 = 0.656/1.07 = 0.613$$

The analysis of the AD statistic confirms and reinforces the findings from the KS and CvM tests.:

- the ratio of the *Log-Normal Test* values to the *Fully Specified Case* values is approximately **0.27-0.33**, providing the **strongest evidence** yet that using standard, distribution-free tables for a test with estimated parameters would be grossly incorrect.
- the pattern observed with the KS and CvM tests holds true: the critical values for the *Log-Normal Test* are again close to those for the *Estimated Normal Test Case*. This consistent finding across all three EDF statistics is a significant result of this study. It strongly suggests that for goodness-of-fit tests where parameters are estimated, the null distributions of the test statistics are primarily determined by the act of parameter estimation itself and are surprisingly insensitive to the transformation (from normal to log-normal) of the underlying distribution.

5. CONCLUSION AND FUTURE WORK:

This paper successfully generated and validated a comprehensive set of critical values for the Kolmogorov-Smirnov (KS), Cramér-von Mises (CvM), and Anderson-Darling (AD) goodness-of-

fit tests for a *log-normal* distribution with estimated parameters.

Through a Monte Carlo simulation and comparison with established benchmarks, this paper demonstrated that the critical values for the log-normal test are nearly identical to those for the normal test when parameters are estimated. This consistency across all three EDF statistics suggests a more general principle: for these tests, the distortion of the null distribution caused by parameter estimation is the dominant factor, overshadowing the influence of the specific underlying distribution's shape (in this case, the transformation from a symmetric normal to a skewed log-normal).

For practitioners, the standard, *distribution-free* critical value tables for a goodness-of-fit test with estimated parameters leads to an overly conservative test with a significant loss of power. However, the widely available tables for the normal test with *estimated parameters* can be used as a reliable and accurate approximation.

With these results, there are avenues for future research. The investigation could be extended to other important distributions, such as the *Weibull* or *Gamma* distributions, to see if this observed insensitivity holds. Furthermore, exploring the impact of different parameter estimation methods (e.g., method of moments) on the critical values could provide deeper insights. Finally, a more extensive *power analysis* (similar to Razali, 2011) of these tests against a wider array of alternative distributions would further clarify their practical performance.

(FUN NOTE ON COMPUTATIONAL PROGRESS)

Lewis, in his foundational 1961 paper on the distribution of the Cramér-von Mises statistic, remarked that determining the full distribution via Monte Carlo methods was "*prohibitive*." He noted that for a sample size of just $n=8$, a simulation of 250,000 replications required **six hours** of computing time on the advanced machines of the era, an IBM 704 Computer

With the computing power nowadays, this can be replicated in probably minutes on a standard laptop running a language like Python, with languages like C++ or Fortran likely even faster. This is not meant to diminish the groundbreaking efforts of these pioneers! On the contrary, it highlights the immense challenges they overcame not too long ago. For contrast, the entire simulation for this paper—including 3 tests, 15 sample sizes (ranging from $n=10$ to 2000), and 10,000 replications for each—was executed on a standard modern laptop using Python in 8.06 seconds, so with 250,000 replications, likely it will take less than 4 minutes.

Imagine what computers can do, give it another 20 years!

REFERENCES

Evans, M., Hastings, N., & Peacock, B. (2000). *Statistical Distributions* (3rd ed.). Wiley.

Lewis, P. A. W. (1961). Distribution of the Cramér-von Mises Statistic. *The Annals of Mathematical Statistics*, 32(3), 759–764.

Lilliefors, H. W. (1967). On the Kolmogorov-Smirnov Test for Normality with Mean and Variance Unknown. *Journal of the American Statistical Association*, 62(318), 399–402.

MathWorks. (n.d.). *Log-normal distribution*. The MathWorks, Inc. Retrieved from <https://www.mathworks.com/help/stats/lognormal-distribution.html>

Razali, N. M., & Wah, Y. B. (2011). Power comparisons of Shapiro-Wilk, Kolmogorov-Smirnov, Lilliefors and Anderson-Darling tests. *Journal of Statistical Modeling and Analytics*, 2(1), 21-33.

Stephens, M. A. (1970). Use of the Kolmogorov-Smirnov, Cramér-Von Mises and Related Statistics Without Extensive Tables. *Journal of the Royal Statistical Society. Series B (Methodological)*, 32(1), 115–122.

Stephens, M. A. (1974). EDF Statistics for Goodness of Fit and Some Comparisons. *Journal of the American Statistical Association*, 69(347), 730–737.

Wolfram. (n.d.). *Log Normal Distribution*. Wolfram MathWorld. Retrieved from <https://mathworld.wolfram.com/LogNormalDistribution.html>

The author acknowledges the use of AI tools in preparing this manuscript. Google's Gemini was utilized for assistance with prose, grammar, and logical structuring. X's Grok provided support for initial research and debugging the Python simulation, and LaTeX code.

APPENDIX

Table A1. Full Critical Values for the Kolmogorov-Smirnov Test (D)

Sample Size (n)	$\alpha = 0.25$	$\alpha = 0.10$	$\alpha = 0.05$	$\alpha = 0.01$
10	0.2093	0.2421	0.2631	0.3037
15	0.1739	0.2014	0.2187	0.2513
20	0.1514	0.1748	0.1910	0.2219
25	0.1377	0.1592	0.1733	0.2015
30	0.1254	0.1454	0.1582	0.1834
40	0.1100	0.1271	0.1384	0.1592
50	0.0990	0.1149	0.1248	0.1450
100	0.0707	0.0816	0.0888	0.1034
200	0.0507	0.0584	0.0634	0.0740
300	0.0412	0.0477	0.0520	0.0605
400	0.0356	0.0410	0.0445	0.0526
500	0.0320	0.0371	0.0403	0.0470
1000	0.0228	0.0263	0.0286	0.0332
1500	0.0185	0.0213	0.0231	0.0268
2000	0.0161	0.0186	0.0203	0.0235

Table A2. Full Critical Values for the Cramér-von Mises Test (W^2)

Sample Size (n)	$\alpha = 0.25$	$\alpha = 0.10$	$\alpha = 0.05$	$\alpha = 0.01$
10	0.0719	0.1003	0.1223	0.1716
15	0.0728	0.1013	0.1220	0.1670
20	0.0719	0.0995	0.1200	0.1682
25	0.0731	0.1014	0.1240	0.1707
30	0.0723	0.1005	0.1223	0.1754
40	0.0732	0.1030	0.1238	0.1727
50	0.0738	0.1033	0.1263	0.1818
100	0.0736	0.1041	0.1242	0.1731
200	0.0736	0.1040	0.1254	0.1796
300	0.0732	0.1042	0.1264	0.1841
400	0.0738	0.1017	0.1224	0.1753
500	0.0741	0.1034	0.1244	0.1798
1000	0.0742	0.1043	0.1263	0.1802
1500	0.0740	0.1017	0.1220	0.1766
2000	0.0743	0.1043	0.1266	0.1826

Table A3. Full Critical Values for the Anderson Darling Test (A^2)

Sample Size (n)	$\alpha = 0.25$	$\alpha = 0.10$	$\alpha = 0.05$	$\alpha = 0.01$
10	0.4364	0.5832	0.6993	0.9594
15	0.4498	0.6015	0.7095	0.9464
20	0.4486	0.5986	0.7085	0.9599
25	0.4579	0.6109	0.7339	0.9828
30	0.4551	0.6050	0.7224	1.0024
40	0.4605	0.6161	0.7322	1.0004
50	0.4645	0.6329	0.7439	1.0342
100	0.4679	0.6305	0.7443	0.9950
200	0.4731	0.6281	0.7470	1.0168
300	0.4648	0.6326	0.7609	1.0609
400	0.4666	0.6204	0.7301	1.0186
500	0.4701	0.6241	0.7417	1.0224
1000	0.4746	0.6358	0.7489	1.0509
1500	0.4687	0.6189	0.7360	1.0154
2000	0.4704	0.6305	0.7514	1.0434